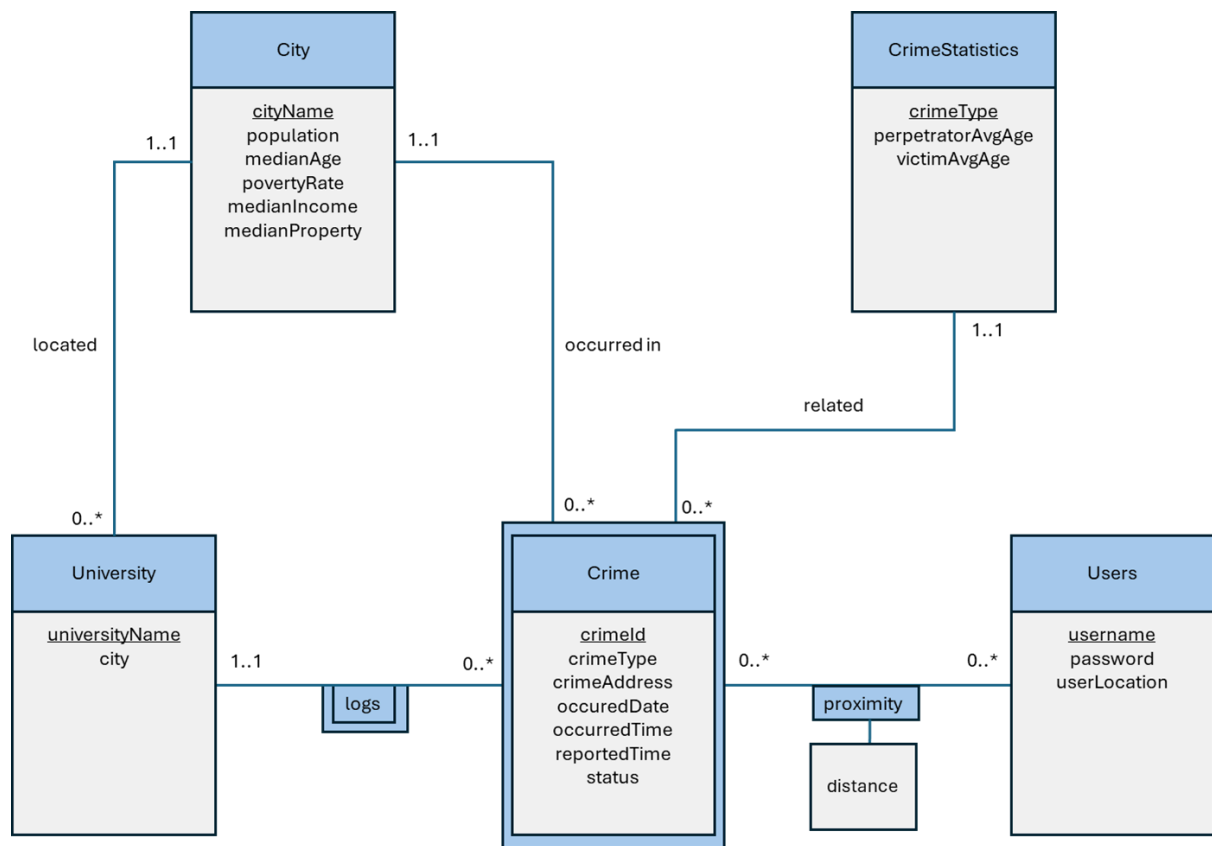




Assumptions

Entities and Attributes

1. Crime:

- Attributes: **crimeId** (PK), **crimeType**, **crimeAddress**, **occuredDate**, **occuredTime**, **reportedTime**, **status**
- Each crime is uniquely identified by its crimeId and the name of the university that reported the crime. It also contains information about its status, location, and type.

2. University:

- Attributes: **universityName** (PK), **city**
- Each university is uniquely identified by its name and states the city it is located in.

3. Users:

- Attributes: **username** (PK), **location_id**(FK), **password**
- Each user is identified by a unique username and contains the location and password.

4. **City:**

- Attributes: `city_name` (PK), `population`, `median age`, `poverty rate`, `median_household_income`, `median property value`
- Each city is uniquely identified by its `cityName` and holds demographic information about people living in the city to identify trends with crime.

5. **Crime Statistics:**

- Attributes: `crimeType` (PK), `perpetratorAvgAge`, `victimAvgAge`
- Each crime type is uniquely identified by its `crimeType` and holds information related that specific crime type across the United States
 - i. For example, the average age of the perpetrator if the `crimeType` was burglary would store the national average age of someone who committed a burglary from 2012-2022

Relationships and Cardinality

1. **University - Crime (Logs) (1, ∞):**

- Each crime occurs at one university
- One university can have multiple crimes

2. **University - City (Located) (∞, 1):**

- Each university can only be located in one city
- Each city can have multiple universities

3. **City - Crime (Occurred In) (1, ∞):**

- Each crime can only occur in one city
- Each city can have multiple crimes

4. **Crime - CrimeStatistics (Related) (∞, 1):**

- Each crime can only have one type
- Each type of crime can have multiple instances

5. **Crime - Users (Proximity) (∞, ∞):**

- Each crime can be close to multiple users
- Each user can be close to the location a crime took place

Normalization:

FDs:

3NF

crimeId-> crimeType,crimeAddress,occurredDate, occurredTime, reporteTime,
status,universityName

universityId->city

username->password,userLocation

cityName->population,medianAge,povertryRate,medianIncome,medianProperty

crimeType->perpetratorAvgAge,victimAvgAge

Crime(crimeId,crimeType,crimeAddress,crimeStatus,universityName)

University(universityName,universityAddress)

Users(userId,username,password,userLocation)

City(cityName, population, medianAge, povertyRate, medianIncome, medianProperty)

CrimeStatistics(crimeType,perpetratorAvgAge,victimAvgAge)

Relational Schema

1. University(universityName VARCHAR(X), city VARCHAR(X), PRIMARY KEY (universityName), FOREIGN KEY city REFERENCES City(cityName))
2. City(cityName VARCHAR(X), population INT, medianAge INT, povertyRate REAL, medianIncome REAL, medianProperty REAL, PRIMARY KEY (cityName))
3. Users(username VARCHAR(X), password VARCHAR(X), userLocation VARCHAR(X), PRIMARY KEY (username))
4. CrimeStatistics(crimeType VARCHAR(X), perpetratorAvgAge INT, victimAvgAge INT, PRIMARY KEY (crimeType))
5. Crime(crimeId INT, crimeType VARCHAR(X), occuredDate DATE, occuredTime VARCHAR(X), reportedTime VARCHAR(X), status VARCHAR(X), universityName VARCHAR(X), city VARCHAR(X), PRIMARY KEY (crimeId, universityName), FOREIGN KEY (universityName) REFERENCES University(universityName) ON DELETE CASCADE, FOREIGN KEY crimeType REFERENCES CrimeStatistics(crimeType), FOREIGN KEY city REFERENCES City(cityName))
6. Proximity(username VARCHAR(X), crimeId INT, distance REAL, PRIMARY KEY (username, crimeId), FOREIGN KEY username REFERENCES Users(username), FOREIGN KEY crimeId REFERENCES Crime(crimeId))